

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1109

Darrah 138 kV
January 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1109. The Interconnection Customer proposes a 211.1 MW solar photovoltaic facility interconnecting at the Darrah 138 kV substation in the West Virginia area. The request was received on November 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 270 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,380,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N MADISON 345 / HURRICANE 138 kV	PENCE / VALMOUNT SUBSTATION 345 kV Line	176.2	279	6.2
W HILL / FIG GARDEN 138 kV L	SIGGEN / HYAMPOM 138 kV	173.4	180	9.8
W HILL / CONTR COSTA 3 138 kV Line	HYAMPOM / SIGGEN 138 kV	155.9	208	9.8
CONTRA COSTA 3 / TRISTATE 138 kV L	SIGGEN / HYAMPOM SUBSTATION 138 kV L	149.0	231	9.8
PONT DES MOUTON / MILTON 138 kV	VALEMOUNT / TRISTATE SUBSTATION 345 kV	147.6	206	9.6
DAVIS RD / HAGAN 138 kV	VALEMOUNT / TRISTATE 345 kV	146.4	238	6.7
HAGAN / SIGGEN 138 kV	VALEMOUNT / TRISTATE 345 kV	145.7	240	6.7
FITZ / PONT DES MOUTON 138 kV	VALEMOUNT / TRISTATE 345 kV Line	145.5	216	9.6
TESORO / BLAND SUBSTATION 138 kV L	VALEMOUNT / TRISTATE 345 kV line	144.3	236	7.5
7TH AVENUE / FITZ 138 kV line	VALEMOUNT / TRISTATE 345 kV	142.1	226	9.6
BOWLES / TESORO 138 kV	VALEMOUNT / TRISTATE 345 kV	141.4	250	7.5
FIG GARDEN SUBSTATION / BOWLES 138 kV	VALEMOUNT / TRISTATE 345 kV line	140.8	253	7.5
DOWELL / WEST PAINTSVILLE 138 kV	HUBBARDSTOWN / TRISTATE 345 kV L	134.2	427	5.1
HUBBARDSTOWN SUBSTATION / BIG SANDY 138 kV	W PAINTSVILLE / BAKER 345 kV	120.2	300	9.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FUEL / DARRAH 138 kV	E HUNTINGTON / DARRAH 138 kV	105.3	186	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH MADISON 345 SUBSTATION / HURRICANE 138 kV	Rebuild line with higher-capacity conductor	\$2,517,000	15% (\$381,000)
WEST HILL / FIG GARDEN 138 kV	Rebuild line with higher-capacity conductor	\$2,445,000	100% (\$2,445,000)
W HILL / CONTR COSTA 3 138 kV Line	Rebuild line with higher-capacity conductor	\$3,876,000	100% (\$3,876,000)
CONTRA COSTA 3 / TRISTATE 138 kV	Rebuild line with higher-capacity conductor	\$6,034,000	100% (\$6,034,000)
PONT DES MOUTON / MILTON 138 kV	Rebuild line with higher-capacity conductor	\$3,893,000	100% (\$3,893,000)
DAVIS RD / HAGAN 138 kV	Rebuild line with higher-capacity conductor	\$17,950,000	100% (\$17,950,000)
HAGAN / SIGGEN 138 kV	Rebuild line with higher-capacity conductor	\$5,036,000	100% (\$5,036,000)
FITZ / PONT DES MOUTON 138 kV	Rebuild line with higher-capacity conductor	\$3,809,000	100% (\$3,809,000)
TESORO / BLAND SUBSTATION 138 kV L	Rebuild line with higher-capacity conductor	\$3,335,000	100% (\$3,335,000)
7TH AVENUE / FITZ 138 kV line	Rebuild line with higher-capacity conductor	\$2,388,000	100% (\$2,388,000)
BOWLES / TESORO 138 kV	Rebuild line with higher-capacity conductor	\$15,626,000	100% (\$15,626,000)
FIG GARDEN SUBSTATION / BOWLES 138 kV	Rebuild line with higher-capacity conductor	\$8,644,000	100% (\$8,644,000)
DOWELL / WST PAINTSVILLE 138 kV	Rebuild line with higher-capacity conductor	\$4,005,000	5% (\$202,000)
HUBBARDSTOWN / BIG SANDY 138 kV	Rebuild line with higher-capacity conductor	\$3,211,000	11% (\$348,000)
FUEL / DARRAH 138 kV	Reconductor line	\$1,413,000	100% (\$1,413,000)
	Total allocated to project		\$75,380,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.